

Order Tracking System Ontology

1. Purpose

The purpose of building this Reference Ontology is to provide a knowledge model for a comprehensive Order Tracking System. It aims to standardize the tracking process from order creation to final delivery, ensuring interoperability between customers, sellers, and couriers.

2. Scope

The ontology focuses strictly on the order lifecycle, logistics, and actor interactions within a tracking system. It covers order details, payment verification, shipment movement, and real-time tracking events.

3. Implementation Language

The ontology has to be implemented in **OWL**

4. Intended End-Users

- **User 1:** Customers wanting to track their packages in real-time.
- **User 2:** E-commerce Admins managing order fulfillment.
- **User 3:** Logistics Managers monitoring courier performance.
- **User 4:** Developers integrating tracking APIs into e-commerce platforms.

5. Intended Uses

- **Use 1:** To provide a semantic structure for tracking data.
- **Use 2:** To automate status updates based on tracking events.
- **Use 3:** To query order history and payment records.
- **Use 4:** To validate shipment routes and delivery timelines.

6. Ontology Requirements

a. Non-Functional Requirements

- **NFR1:** The ontology must be consistent and able to answer all competency questions
- **NFR2:** The ontology must follow **FAIR principles** (Findable, Accessible, Interoperable, Reusable) by using standard URIs like Schema.org and FOAF.

b. Functional Requirements: Groups of Competency Questions

Group	Competency Questions (CQs)
Group 1: Actors	CQ1. Who is the customer who placed Order X? CQ2. Which courier company is responsible for Shipment Y? CQ3. Who is the admin that approved the refund?
Group 2: Order Status	CQ1. What is the current status of Order X? CQ2. List all orders that are currently "In Transit". CQ3. Which orders were cancelled before shipment?
Group 3: Logistics	CQ1. What is the tracking number for Shipment Z? CQ2. At what time did the package arrive at the "Sargodha Hub"? CQ3. What is the estimated delivery date for Order X?
Group 4: Payments	CQ1. Was Order X paid via Credit Card or Cash on Delivery? CQ2. Which orders have a "Failed" payment status? CQ3. What is the total invoice amount for Customer A?
Group 5: Catalog	CQ1. Which products are included in Order X? CQ2. What is the quantity of "Laptops" in Shipment Y? CQ3. Which seller provided Product Z?
Group 6: Exceptions	CQ1. Which shipments are currently "Delayed"? CQ2. What is the reason for the Return Request of Order X? CQ3. List all orders that were returned due to damage.

7. Pre-Glossary of Terms (Based on CQs)

Terms from Competency Questions:

- **Actors:** Customer, Seller, Courier, Admin.
- **Transactions:** Order, Payment, Invoice, ReturnRequest, Cancellation.

- **Logistics:** Shipment, TrackingEvent, ShippingAddress, TrackingNumber, Route.
- **Catalog:** Product, OrderItem, Category, UnitPrice.
- **Statuses:** OrderStatus (Pending, Shipped, Delivered), PaymentStatus (Paid, Unpaid).